North Bengal International University

Lab Report: 04

Exp Name : A program to short an array using Bubble short in c program .

Subject : Data Structure

Subject Code : CSE 0612-2131

Submitted By:

Name : Md. Tanvir Rahman Pranto
Dept : CSE
Roll : 0712340012101005
Semester : 2nd Year, 4th semester.

Submitted To:

Name : Saifur Rahman
Designation : Lecturer
Dept : CSE
Date :

Code:

```c
#include <stdio.h>

int main() {
    int a[6] = {5, 2, 9, 1, 6, 3};
    int n = 6, i, j, temp;

    printf("Before sorting: ");
    for (i = 0; i < n; i++) {
        printf("%d ", a[i]);
    }
    printf("\n");

    for (i = 0; i < n - 1; i++) {
        for (j = 0; j < n - i - 1; j++) {
            if (a[j] > a[j + 1]) {
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }

    printf("After sorting: ");
    for (i = 0; i < n; i++) {
        printf("%d ", a[i]);
    }
    printf("\n");

    return 0;
}
```

Output:

```
OUTPUT    TERMINAL

Before sorting: 5 2 9 1 6 3
After sorting: 1 2 3 5 6 9
```

Conclusion: This program sorts an array using **Bubble Sort**, repeatedly swapping adjacent elements until the array is in order. It's simple, effective, and great for small datasets

a[j]	a[j+1]	a[j] > a[j+1]	Swap
5	2	✅ Yes	{2, 5, 9, 1, 6, 3}
5	9	❌ No	{2, 5, 9, 1, 6, 3}
9	1	✅ Yes	{2, 5, 1, 9, 6, 3}
9	6	✅ Yes	{2, 5, 1, 6, 9, 3}
9	3	✅ Yes	{2, 5, 1, 6, 3, 9}
2	5	❌ No	{2, 5, 1, 6, 3, 9}
5	1	✅ Yes	{2, 1, 5, 6, 3, 9}
5	6	❌ No	{2, 1, 5, 6, 3, 9}
6	3	✅ Yes	{2, 1, 5, 3, 6, 9}
2	1	✅ Yes	{1, 2, 5, 3, 6, 9}
2	5	❌ No	{1, 2, 5, 3, 6, 9}
5	3	✅ Yes	{1, 2, 3, 5, 6, 9}
1	2	❌ No	{1, 2, 3, 5, 6, 9}
2	3	❌ No	{1, 2, 3, 5, 6, 9}

Final sorted array: **1 2 3 5 6 9**